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An Alternate Marking Deployment Framework
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Abstract

This document provides a framework for Alternate Marking deployment and includes considerations and guidance for the deployment of the Alternate Marking methodology.

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1. Introduction

The Alternate Marking [RFC9341] and Clustered Alternate Marking [RFC9342] define the Alternate Marking technique that is a hybrid performance measurement method, per [RFC7799] classification of measurement methods. This method is based on marking consecutive batches of packets and it can be used to measure packet loss, latency, and jitter on live traffic.

The first experiments on Alternate-Marking were described in [RFC8321] and [RFC8889].

According to the definitions of [RFC7799], the Alternate-Marking Method can be classified as Hybrid Type I. Indeed, Alternate Marking can be implemented by using reserved bits in the protocol header, and the change in value of these marking bits at the source node is formally considered a modification of the stream of interest.

This document complements [RFC9341] and [RFC9342] as it explains the mechanisms that can be used to manage and deploy the method.

This document can be considered as the base document to deploy the Alternate Marking Method. There are three complementary documents to be considered for the manageability aspects:

[I-D.ietf-bpm-alt-mark-yang], ~~that~~ which defines a YANG Data Model for the Alternate Marking Method for configuration;

[I-D.ietf-bpm-on-path-telemetry-yang], ~~that~~ which defines an On-Path Telemetry YANG Data Model for notification;

[I-D.ietf-opsawg-ipfix-alt-mark], ~~that~~ which defines IPFIX Alternate-Marking Information Elements.

1.1. Requirements Language

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

1.2. Terminology

Abbreviations used in this document:

AltMark: Alternate-Marking

NMS: Network Management System

IPv6: Internet Protocol version 6

SRv6: Segment Routing over IPv6 data plane~~dataplane~~

BIER: Bit Index Explicit Replication

MPLS: Multi-Protocol Label Switching

SFC: Service Function Chaining

NVO3: Network Virtualization Overlays

IPFIX: IP Flow Information Export

~~YANG: Yet Another Next Generation~~

PCEP: Path Computation Element Communication Protocol

~~BGP: Border Gateway Protocol~~

2. AltMark Deployment Domain

The Alternate Marking Method MUST be deployed in a controlled domain for security and compatibility reasons. In this regard, [RFC8799] reports further examples of specific limited domain solutions. It is not common that the user traffic originates and terminates within the controlled domain. For this reason, it will typically only be applicable in an overlay network, where user traffic is encapsulated at one domain border, decapsulated at the other domain border and the encapsulation incorporates the relevant extension header for Alternate Marking. This requirement also implies that an implementation MUST filter packets that carry Alternate Marking data and are entering or leaving the controlled domain.

A controlled domain is a managed network where it is required to select, monitor and control the access to the network by enforcing policies at the domain boundaries in order to discard undesired external packets entering the domain and check the internal packets leaving the domain. It does not necessarily mean that a controlled domain is a single administrative domain or a single organization. A controlled domain can correspond to a single administrative domain or can be composed by multiple administrative domains under a defined network management. Indeed, some scenarios may imply that the Alternate Marking Method involves more than one domain, but in these cases, it is RECOMMENDED that the multiple domains create a whole controlled domain while traversing the external domain by employing

Commenté [MB1]: Do not expand per https://rpc-wiki.rfc-editor.org/rpc/wiki/doku.php?id=abbrev_list

Commenté [MB2]: No need to expand per https://rpc-wiki.rfc-editor.org/rpc/wiki/doku.php?id=abbrev_list

Commenté [MB3]: Why it MUST be deployed by any random network?

I guess you meant, if deployed, it MUST be deployed in controlled domain?

Commenté [MB4]: What is meant by implem here? Is there a border node altmark implem? Else?

Commenté [MB5]: Or «be able to filter» or «filter by default»?

Commenté [MB6]: Alt marking may be in several layers. Also require that filtering be supported in all layers.

IPsec authentication and encryption or other VPN technology that provides full packet confidentiality and integrity protection. ~~In a few words, i~~It must be possible to control the domain boundaries and eventually use specific precautions if the traffic traverse the Internet. Note that, if there is a trust relationship between different entities, the controlled domain can still be considered. The different entities just need to agree on how to manage it in terms of configuration and reporting.

The Alternate Marking measurement domain can overlap with the controlled domain or may be a subset of the controlled domain. The typical scenarios for the application of the Alternate Marking Method depend on the controlled domain boundaries, in particular:

* the user equipment can be the starting or ending node, only in case it is fully managed and if it belongs to the controlled domain. In this case the user generated packets contain the Alternate Marking data. But, in practice, this is not common due to the fact that the user equipment cannot be totally secured in the majority of cases.

* the ~~CPE (~~Customer Premises Equipment (CPE) or the ~~PE (~~Provider Edge (PE)
routers are most likely to be the starting or ending nodes since they can be border routers of the controlled domain. For instance, the CPE, which connects the user's premises with the service provider's network, belongs to a controlled domain only if it is managed by the service provider and if additional security measures are taken to keep it trustworthy. Typically, ~~the~~ a CPE or ~~the~~ PE can encapsulate a received packet in an outer header which contains the Alternate Marking data. They can also be able to filter and drop packets from outside of the domain with inconsistent fields to make effective the relevant security rules at the domain boundaries, for example, ~~for example~~ a simple security check can
be to insert the Alternate Marking data if and only if the destination is within the controlled domain.

3. AltMark Measurement Nodes

An Alternate-Marking Domain consists of marking nodes, unmarking nodes, and transit nodes.

~~-~~A marking node, also called encapsulating node, incorporates the AltMark Data Fields into packets in order to enable Alternate-Marking. If ~~the~~ an Alternate-Marking method is enabled for a selected flow of the traffic, the encapsulating node is responsible for applying the AltMark functionality to the selected flow and to take initial timestamps and packet counters.

Commenté [MB7]: Fix the formatting

Idem for the subsequent paragraphs.

Commenté [MB8]: Should we mention the behavior of a ingress node if it receives a packet with altmarking? I guess default behavior is to ignore (strip?) it?

Commenté [MB9]: To be defined?

Commenté [MB10]: As there might be more (eventually, at distinct layers)

Commenté [MB11]: That is?

A transit node only reads AltMark Data Fields in order to take timestamps and packet counters.

An unmarking node, also called decapsulating node, reads AltMark Data Fields in order to take final timestamps and packet counters and then removes any AltMark Data Fields from packets.

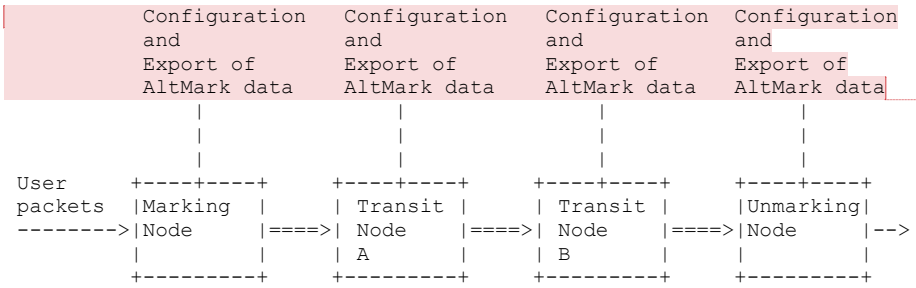


Figure 1: Roles of Alternate-Marking Nodes

4. Type of Measurements

The methodology described in the previous sections can be applied to various performance measurement problems. The only requirement is to select and mark the flow to be monitored~~; . in~~ In ~~this way~~doing so, packets are batched by the sender, and each batch is alternately marked such that it can be easily recognized by the receiver.

Either one or two flag bits might be available for marking in different deployments:

One flag: packet loss measurement MUST be done as described in Section 3.1 of [RFC9341], while delay measurement MUST be done according to the single-marking method described in Section 3.2.1 of [RFC9341]. Mean delay (Section 3.2.1.1 of [RFC9341]) MAY also be used but it could imply more computational load.

Two flags: packet loss measurement MUST be done as described in Section 3.1 of [RFC9341], while delay measurement MUST be done according to double-marking method Section 3.2.2 of [RFC9341]. In this case~~,~~ single-marking MAY also be used in combination with double-marking and the two approaches provide slightly different pieces of information that can be combined to have a more robust data set.

Commenté [MB12]: Maybe worth clarifying whether (o rnot) all internal nodes have to be altmark aware

Commenté [MB13]: Can we be more specific or at least add a mention about which counters we are talking about?

Commenté [MB14]: Also need to expose altmark capabilities. See also below

There are some operational guidelines to consider for the purpose of deciding to follow the recommendations above and use one or two flags:.

The Alternate-Marking method utilizes specific flags in the packet header, so an important factor is the number of flags available for the implementation. Indeed, if there is only one flag available there is no other way, while if two flags are available the option with two flags is certainly more complete.

The duration of the Alternate-Marking period affects the frequency of the measurement and this is a parameter that can be decided on the basis of the required temporal sampling. But However, it cannot be freely chosen, as explained in Section 5 of [RFC9341].

The Alternate-Marking methodologies enable packet loss, delay, and and delay variation calculation, but in accordance with the method used (e.g., e.g. single-marking or double-marking), there are different kinds of information that can be derived. For example, to get more statistics of extent data, the option with two flags is desirable. For this reason, the type of data needed in the specific scenario is an additional element to take into account.

The Alternate-Marking methods imply different computational load depending on the method employed. Therefore, the available computational resources on the measurement points can also influence the choice. As an example, mean delay calculation may require more processing and it may not be the best option to minimize the computational load.

A deployment of the Alternate-Marking Method should also take into account how to handle and recognize marked and unmarked traffic. Since Alternate-Marking normally employs a marking field which is dedicated, reserved, and included in a protocol extension, the measurement points can learn whether the measurement is activated or not by checking if the specific extension is included or not within the packets.

5. An AltMark Deployment Framework

Figure 2 shows an overview of the Alternate-Marking Deployment Framework, including Network Configuration and Data Collection functions, which can also be implemented in different systems.

Commenté [MB15]: Outer packet header?

Commenté [MB16]: This need to be highlighted. See my comment about capabilities.

Commenté [MB17]: Not sure this is worth a section. At least some elaboration will be needed.

BTW, as multiple altmark schemes can be supported, we need at least support of the following:

- Means for nodes to expose their altmark capabilities
- Means to control the layer/scheme used for altmark
- Checks that a domain is consistentely configured

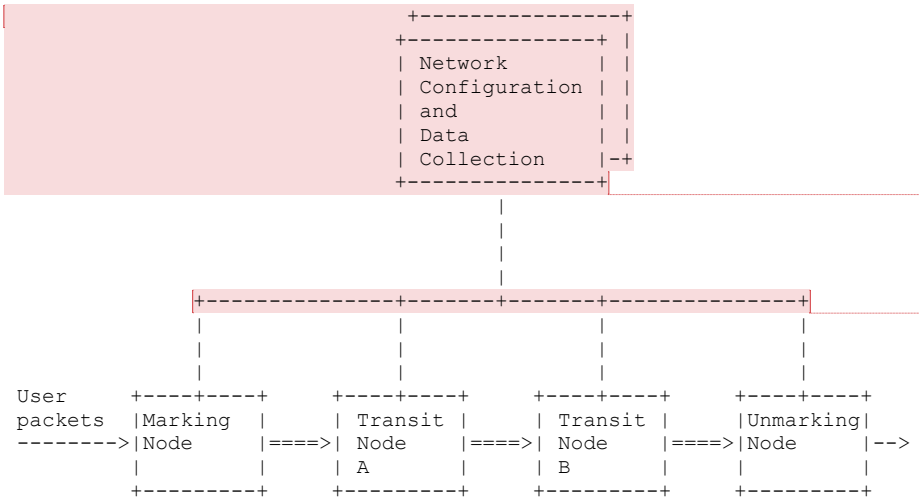


Figure 2: Alternate-Marking Framework with Configuration and Data Export

6. Configuration

Several mechanisms can be used for configuration and, in particular, YANG-based management protocol ~~Model~~, PCEP, and BGP. These are needed to signal and configure the parameters to identify the flow to monitor both in case of point-to-point flow and multipoint-to-multipoint flow. Indeed, the selection of the identification fields directly affects the type of paths that the flow would follow in the network (e.g., ~~e.g.~~ a 5-tuple can usually be used to identify a monitored flow).

For IPv6, the setting of the Flow Monitoring Identification (FlowMonID) is used to identify a flow, also in combination with other parameters, like source and destination addresses, as described in [RFC9343]. FlowMonID can be pseudo-randomly generated by the source node or assigned by the central controller. It can be algorithmically generated by the source node, which can set it pseudo-randomly with some chance of collision, but this approach cannot guarantee the uniqueness of FlowMonID since conflicts and collisions are possible. It ~~would be~~is ~~advisable~~ RECOMMENDED that the FlowMonID is assigned by the central controller. Indeed, since the controller knows the network topology, it can allocate the value properly to avoid or minimize ambiguity and guarantee ~~the~~ uniqueness. Further details can be found in section 5.3 of [RFC9343].

Commenté [MB18]: As «controller» is mentioned in the doc. Maybe consider a mention that the same or distinct controllers may be used for this.

Commenté [MB19]: Need some words about discovery of nodes and provisioning tasks to make sure that a node is talking with an authorized controller.

Commenté [MB20]: Before configuring, there is a need to assess/determine/retrieve first the almmark capabilities of a node.

Commenté [MB21]: Add references

Additionally, other parameters are essential for the activation of the AltMark methodology: the choice between end-to-end or hop-by-hop measurement, the choice between the methods with one flag or two flags and the duration of the Alternate-Marking period which affects the measurement frequency (longer the duration of the block, the less frequently the measurement can be taken).

6.1. YANG Data Model

~~The~~ A YANG data model can be used for the definition of the AltMark data sent over network management protocols such as the NETCONF and RESTCONF. They can be used for configuring Alternate-Marking in network nodes that support it. An example of ~~the~~ an Alternate-Marking YANG data model is defined in [I-D.ietf-bpm-alt-mark-yang].

6.2. PCEP and BGP

There are also other control plane mechanisms to advertise and activate AltMark capabilities, using PCEP and BGP: [I-D.ietf-idr-sr-policy-ifit], [I-D.ietf-idr-bgp-ifit-capabilities], [I-D.ietf-pce-pcep-ifit], and [I-D.he-idr-bgp-flowspec-ifit].

Commenté [MB22]: Say whether both are needed

7. Data Export

Each packet marked for Alternate-Marking~~,~~ as~~—~~, for example, the AltMark IPv6 option type defined in Section 3.1 of [RFC9343] or the Segment Routing TLV Type as defined in Section 3.1 of [RFC9947] is counted or timestamped according to the method. The export of data can be done using the IPFIX or YANG push metering process depending which Network Telemetry [RFC9232] protocol is used to export the data.

There are at least two options for the computation:

(1) A centralized solution where the NMS acts as a Collector. The loss and delay metrics are computed on the NMS by comparing the packet counts and timestamps at each AltMark period, according to the AltMark technique [RFC9341] [RFC9342].

Commenté [MB23]: Expand

(2) A distributed solution where the computation of loss and delay metrics can be done directly on the node. In this case, the relevant counters and timestamps must be available on the node and can be carried with the packet or can be transmitted by introducing a new protocol or by extending existing protocols. For example, according to [RFC9951], it is possible to export the delay performance metrics in IPFIX. Therefore, assuming we have the counter and timestamp information directly on the node, as enabled by [RFC9947], the delay can be computed on the router.

When data are collected, packet counts and timestamps are reported to the collector, but a certain synchronization mechanism is required to ensure that the collected data is correlated properly. Therefore, the Period Number (PN) can be used to help to determine the packet counts related to the same block of markers, or the timestamps related to the same marked packet. The PN is generated each time a node reads the packet counts or timestamps, and is associated with each packet count and timestamp reported. ~~The assumption is that the~~ nodes are REQUIRED to be ~~time~~ time-synchronized as described in [RFC9341] and [RFC9342].

The PN can be calculated by dividing the local time (when the counts or timestamps are read) by the interval of the marking time period and then rounding down to the nearest whole integer using the floor function. This calculation does not require strict time synchronization, since the synchronization required for AltMark to function is enough, and the PN value will be always the same on each node under these conditions. In addition, if the PN calculation is resource consuming for the node, it may also be calculated on the NMS since the timestamp information is available there.

Note that, in addition to IPFIX, YANG-Push and Hybrid Two-Step, which are reported below, it is also possible to ~~employ~~ use gRPC, an open source Remote Procedure Call (RPC) framework for exporting telemetry data.

7.1. IPFIX

For IPFIX [RFC7011], new Information Elements (IEs) for FlowMonID, Loss and Delay ~~flag~~ flags are needed so that the data can be aggregated according to the Alternate-Marking Method. The new IEs to export Alternate Marking measurement data are specified in [I-D.ietf-opsawg-ipfix-alt-mark].

The IPFIX entities, which are of interest to describe the relationship to the forwarding topology and the control~~-plane~~ plane are further described in [I-D.ietf-opsawg-ipfix-alt-mark].

7.2. YANG-Push

Commenté [MB24]: Can these co-exist? Of only one is needed in a deployment?

~~For YANG-Push [RFC8641],~~ P ~~p~~eriodic subscription as defined in Section 3.1 of [RFC8641] is used to subscribe data. [I-D.ietf-bpm-on-path-telemetry-yang] introduces a YANG data model for monitoring Alternate-Marking telemetry data.

Since the amount of observed data could overwhelm a router processor on a network node, publishing data from network processors as specified in [I-D.ietf-netconf-distributed-notif] is advised.

7.3. Hybrid Two-Step

Another option for collecting AltMark data can be the Hybrid Two-Step (HTS) approach, defined in [I-D.ietf-ippm-hybrid-two-step]. It is a method of telemetry collection that separates the act of measuring the performance metric from collecting this information. A packet in the flow to which the AltMark method is applied can be used as an HTS Trigger, while the HTS Follow-up packet is then generated to collect the measurement data from the nodes, as further detailed in [I-D.ietf-ippm-hybrid-two-step].

8. Encapsulations

8.1. IPv6

The Alternate-Marking encapsulation for IPv6 is defined in [RFC9343], which also discusses deployment considerations for IPv6 networks.

The IPv6 AltMark Option [RFC9343] applies the Alternate Marking Method to IPv6, and defines an Extension Header Option to encode the Alternate Marking Method for both the Hop-by-Hop Options Header and the Destination Options Header.

For example, [I-D.zhou-ippm-enhanced-alternate-marking] defines extended data fields for the IPv6 AltMark Option and provides enhanced capabilities to overcome some challenges and enable future proof applications. It is worth mentioning that the enhanced capabilities are intended for further use and are optional.

Additional extended data fields for the IPv6 AltMark Option are introduced in [I-D.wang-ippm-ipv6-flow-measurement] and [I-D.wang-ippm-ipv6-distributed-flow-measurement].

8.2. SRv6

The Alternate-Marking encapsulation for SRv6 is discussed in IPv6 AltMark Option [RFC9343] and [RFC9947]. However, it is to be noted that [RFC9343] also applies to SRv6. ~~While~~ [RFC9947] is an alternative experimental approach and it uses an experimental TLV in the SRH.

8.3. BIER

The Alternate-Marking encapsulation for BIER is introduced in [I-D.ietf-bier-pmmm-oam].

Commenté [MB25]: When this option is worth to use compared to the other methods?

8.4. MPLS

The Alternate-Marking application to MPLS has some options, as introduced in [RFC9571], [RFC9714], or [I-D.cx-mpls-mna-inband-pm].

8.5. SFC

The Alternate-Marking encapsulation for SFC is introduced in [I-D.mfm-ippm-sfc-nsh-pmamm].

Commenté [MB26]: Expired since 2022. Do we need to keep it?

8.6. NVO3

The Alternate-Marking encapsulation for NVO3 is introduced in [I-D.fmm-nvo3-pm-alt-mark].

Commenté [MB27]: Expired since 2019. Do we need to keep this?

9. Operational Considerations

This document aims to complement [RFC9341] and [RFC9342] in terms of operation and manageability of the methodology. Indeed, the previous sections cover the aspects related to configuration and data export.

[RFC9341] and [RFC9342] already include some recommendations for deployment. Further details about the method operation are specified for the different extensions: IPv6 [RFC9343], SRH [RFC9947], and MPLS [RFC9714].

In a controlled domain, the nodes may support the AltMark specific encapsulation and this also depends on the implementation. If a node is configured to read the AltMark option, the measurement is done on that node, otherwise, it is simply not considered in the measurement.

Assuming that the measurement domain overlaps with the controlled domain, the procedure for AltMark data encapsulation can be summarized as follows:

* Ingress Marking Node: ~~the~~ an Ingress Node of a controlled domain that supports the Alternate Marking Method adds the AltMark data in the ~~the~~ data packets.

Commenté [MB28]: As there might be many

* Intermediate Transit Node: if an Intermediate Node is not capable of processing the AltMark data, it simply ignores it. If an Intermediate Node is capable of processing the AltMark data, it processes it.

* Egress Unmarking Node: ~~The~~ An Egress Node is the last node of the controlled domain. The processing if the AltMark data is similar to the processing at the Intermediate Nodes. The only difference is that it needs by default to remove the AltMark data from the ~~the~~ data packets.

All the nodes in the AltMark measurement domain can be configured using the YANG data model in [I-D.ietf-bpm-alt-mark-yang], while the data can be exported using IPFIX [I-D.ietf-opsawg-ipfix-alt-mark] or the on-path telemetry YANG model [I-D.ietf-bpm-on-path-telemetry-yang].

Commenté [MB29]: To cover the case of multiple network/same admin domain you mentionned elsewhere in the doc.

Commenté [MB30]: Why «all»?

9.1. Measurement Domain Considerations

As mentioned in Section 2, the AltMark measurement domain can overlap with the controlled domain or may be a subset of the controlled domain.

In general, the flow identification parameters and flow correlation values are valid only within a measurement domain.

The uniqueness of the FlowMonID MUST be guaranteed only inside the measurement domain, therefore separate measurement domains may use the same FlowMonID to identify different flows. However, an implementation may use globally unique FlowMonID values across different measurement domains in order to use the same identification fields for a flow that crosses separate measurement domains. This approach could be chosen to simplify the data analysis. The PN is univocally defined in Section 7, but its value may vary in different domains depending on the length of the marking period selected for each domain. ~~Anyway~~Overall, the same marking period SHOULD be used in different measurement domains if the measurement data need to be linked for analysis purposes.

It is RECOMMENDED to decide a priori the measurement domain or the set of measurement domains in which the identifiers are guaranteed unique and the values are meaningful. In this way, ~~the~~ an NMS can handle ~~properly~~ adequately both the configuration and the correlation of the flow identification parameters.

9.2. Synchronization and Timing Considerations

The requirement of time synchronization is defined in S~~s~~ection 5 of [RFC9341] and ~~section~~ Section 8 of [RFC9342]. However, it ~~would be~~is worth considering what happens in case the requirement of time synchronization is not met. If the clock skew or drift implies that the synchronization and timing precondition cannot be satisfied, the measurement must be skipped for the affected intervals until synchronization is restored. The synchronization and timing issue

may affect only a single node or the entire AltMark measurement domain.

If the issue affects only a single node, the measurement can still be done on all the other nodes.

If the issue affects multiple nodes (including the marking nodes) in the measurement domain, the measurement can restart as soon as the issue is resolved.

9.3. Configuration and Data Export Considerations

~~It is worth considering that~~F~~ f~~ault conditions may affect configuration (Section 6) and data export (Section 7) reliability. There are different possibilities:

* Configuration failures or missing exports may involve only a single node that is not a marking node. In this case the measurement can still be done for all the other nodes.

* A configuration failure may involve multiple nodes and, if the marking node(s) is not configured properly, the measurement cannot be done on the other nodes. The measurement can start when the configuration succeeds.

* The missing exports may involve one or more consecutive marking periods on multiple nodes and, in this case, if the marking is still working on the source node(s), the measurement is skipped only on the affected nodes and can continue from the next interval when the exports resume.

~~Note that c~~Configuration mistakes and erroneous values exported can also be treated respectively as configuration failures and missing exports. But, an implementation may consider different workarounds according to the situation.

10. Security Considerations

Alternate Marking [RFC9341] and Clustered Alternate Marking [RFC9342] analyze different security concerns and related solutions. These aspects are valid and applicable also to this document. In particular the fundamental security requirement is that Alternate Marking MUST only be applied in a specific limited domain, as also mentioned in [RFC8799].

Commenté [MB31]: Fix formatting

Commenté [MB32]: Fix formatting

Commenté [MB33]: We need to cover at least the security requirements for the interface between collector/controller and underlying nodes.

Also, highlight that revealing exported data to unauthorilzied enities should be prevented. Access to that data must be secured, etc.

Means to monitor and detect misbehaving nodes within a network should be put in place by an operator.

11. IANA Considerations

This document has no request to IANA.

12. Acknowledgements

The authors of this document would like to thank Greg Mirsky, Chongfeng Xie, Xiao Min, ~~Benoit~~ Benoît Claise, Saumya Dikshit for their comments and reviews.

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